MILITARY SPECIFICATION

FREQUENCY METERS, COAXIAL:
FR-153/USM-129, FR-154/USM-129, FR-155/USM-129

1. SCOPE

1.1 Scope. - This specification covers a series of direct reading coaxial cavity frequency meters measuring frequency in the range from 0.95 kilomegacycles per second (kmcps) to 8.2 kmcps. These frequency meters form part of Microwave Calibration Test Sets AN/USM-129, AN/USM-129A and AN/USM-129B.

1.2 Classification. - The frequency meters covering the frequency range of 0.95 to 8.2 kmcps shall be designated by the Ordnance Part Numbers, given in Table I.

TABLE I - FREQUENCY METERS - DESIGNATION

Frequency Range (kmcps)	Designator	Ordnance Part Number
0.95 to 2.0	FR-153/USM-129	10518881
1.90 to 4.0	FR-154/USM-129	10518906
3.95 to 8.2	FR-155/USM-129	10518835

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-I-45208 Inspection Requirements; General Specification for

DRAWINGS

Ordnance Corps

D10518835 Frequency Meter, Coaxial 3.95 to 8.0 kmcps
D10518881 Frequency Meter, Coaxial 0.95 to 2.0 kmcps
D10518906 Frequency Meter, Coaxial 1.90 to 4.0 kmcps

OTHER PUBLICATIONS Consolidated Freight Classification Rules

FSC 1290

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by contracting officer.)

3. REQUIREMENTS

3.1 Initial Production Sample. - Unless otherwise directed by the contracting officer, an initial production sample shall be required. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production.

3.2 Materials. - Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction. - The frequency meters shall be manufactured in accordance with drawings D10518881, D10518906 or D10518835, as required, and all drawings referenced thereon. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired electrical and mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities.

3.4 Electrical.

3.4.1 Type. - The frequency meters shall be of the absorbtion type.

3.4.2 Voltage Standing Wave Ratio (VSWR). - The maximum VSWR (when the meter is at the operating frequency) shall be 1.5.

3.4.3 Dip. - The minimum percentage dip of the frequency meters shall be as shown in Table II.

TABLE II - PERCENTAGE DIP (MIN.)

Ordnance Part Number	Dip
10518881	20
10518906	25
10518835	30

3.4.4 Q. - The minimum Q shall be one thousand.

3.4.5 Accuracy. - The accuracy within the temperature range of 5°F to 70° F shall be 0.2%.

3.5 Environmental

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3.5.1 Operating.

3.5.1.1 Temperature. - Prior to, during and after being exposed and thermally stabilized at ambient temperatures of plus 32 degrees and plus 110 degrees Fahrenheit, the frequency meter shall meet the requirements of 3.4.

3.5.1.2 Humidity. - Prior to, during and after being subjected to the procedure specified in 4.7.1.2, the frequency meter shall meet the requirements of 3.4.

3.5.1.3 Altitude. - The frequency meter shall be capable of operating at a reduced pressure of 20.58 inches of mercury (equal to an altitutde of 10,000 feet) for four hours.

3.5.2 Non-Operating. -

3.5.2.1 Temperature. - Prior to, and after being subjected to the procedures specified in 4.7.2.1, the frequency meter shall meet the requirements of 3.4.

3.5.2.2 Altitude. - Prior to, and after being subjected to the procedures specified in 4.7.2.2, the frequency meter shall meet the requirements of 3.4.

3.5.2.3 Bench Handling. - After being subjected to the procedure specified in 4.7.2.3, the frequency meter shall show no physical damage and shall meet the requirements of 3.4.

3.5.2.4 Vibration. - After being subjected to the procedure specified in 4.7.2.4, the frequency meter shall show no physical damage and shall meet the requirements of 3.4.

3.5.2.5 Shock. - After being subjected to the procedure specified in 4.7.2.5, the frequency meter shall show no physical damage and shall meet the requirements of 3.4.

3.6 Workmanship. - The equipment, including all parts and accessories, shall be manufactured and finished in a thoroughly workmanlike manner and in accordance with good commercial practice.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilities. - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 General Requirements. - Inspection shall be in accordance with Specification MIL-I-45208.

4.3 Initial Production Sample. - Prior to production, the contractor shall submit the initial production sample in accordance with 3.1 for inspection by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.3.1 Production Sample Failure. - Should the production sample fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

4.4 Material. - Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

4.5 Inspection of Components. - The contractor shall ascertain that prior to assembly all components procured under separate specifications or drawings, have been inspected, tested and accepted in accordance with their respective specifications or drawings.

4.6 Electrical.

4.6.1 Voltage Standing Wave Ratio (VSWR). - The VSWR shall be measured by the slotted section technique. The frequency meter shall be terminated by a matched load of 1.03 VSWR maximum. The value of VSWR shall comply with 3.4.2 at each test frequency given in 4.6.5.

4.6.2 Dip. - Using a suitable cathode ray oscilloscope, the resonant dip of the frequency meter shall be superimposed on a mode curve. The ratio of the height of the dip to the height of the mode curve when the center of the dip coincides with the top of the mode curve is the percentage dip. The percent dip shall comply with 3.4.3 at each test frequency in 4.6.5.

4.6.3 Q. - Using a dual trace oscilloscope, a double display of the mode curve of a suitable oscillator, shall be obtained. The frequency meter under test, shall be tuned so that its dip as presented on one of the mode traces coincides with the top of the second mode trace. A calibrated frequency marker shall be set at the center frequency of the dip (f_c). The frequency marker shall be tuned to the half-power points (f_1, f_2) of the dip trace. The quantity, Q, shall be defined as

$$Q = \frac{f_c}{f_2 - f_1}$$

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The value of Q shall comply with 3.4.4 at each test frequency given in 4.6.5.

4.6.4 Accuracy. - Using a dual trace oscilloscope, a double display of the mode curve of a suitable oscillator shall be obtained. At each of the test frequencies in 4.6.5, a calibrated frequency marker (f_o) shall be set on one of the mode traces. The frequency meter under test shall be tuned so that the center of its dip as displayed on the second mode curve coincides with the frequency marker. The dial reading (f_m) of the frequency meter shall be recorded. The error of the frequency meter shall be defined as

$$\text{Percent error} = \frac{f_o - f_m}{f_o} \times 100$$

The error shall not exceed the value given in 3.4.5.

4.6.5 Test Frequencies. - The test frequencies for VSWR, dip, Q and accuracy shall be those in Table III.

TABLE III - TEST FREQUENCIES (mcps)

Parameter	Ordnance Part Number		
	10518881	10518906	10518835
VSWR, Dip and Q	950 1250 1500 1750 2000	1900 2500 3000 3500 4000	3950 From 4200 to 8200, every 200 mcps.
Accuracy	From 950 to 2000, every 50 mcps	From 1900 to 4000, every 100 mcps	From 3950 to 8200, every 200 mcps

4.7 Environmental.

4.7.1 Operating

4.7.1.1 Temperature. - When tested while exposed and thermally stabilized at ambient temperatures of plus 32 degrees and plus 110 degrees Fahrenheit, the frequency meter shall conform to the requirements of 3.4.

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4.7.1.2 Humidity. - After having been exposed to an atmosphere of 95 percent relative humidity at plus 77 degrees Fahrenheit for 24 hours in non-operating condition, the frequency meter, when tested while exposed to the same temperature and relative humidity, shall comply with the requirements of 3.4 for eight hours of continuous operation.

4.7.1.3 Altitude. - When tested while exposed to a reduced pressure of 20.58 inches of mercury (equal to an altitude of 10,000 feet), the frequency meter shall comply with the requirements of 3.4 for four hours of continuous operation.

4.7.2 Non-Operating

4.7.2.1 Temperature. - After having been stored for four hours while exposed and thermally stabilized at an ambient temperature of minus 25 degrees Fahrenheit, the frequency meter shall be returned to normal conditions (plus 77 degrees Fahrenheit, relative humidity 60 percent, sea level) and tested for compliance with 3.4. The test shall be repeated at an ambient temperature of plus 158 degrees Fahrenheit.

4.7.2.2 Altitude. - After having been stored for one hour while exposed to a reduced pressure of 3.44 inches of mercury (equal to an altitude of 50,000 feet), the frequency meter shall be returned to normal conditions and tested for compliance with 3.4.

4.7.2.3 Bench Handling. - On a hardwood table, in its normal operating position, the frequency meter shall be lifted on one edge 30 degrees from the horizontal or a maximum of four inches, whichever is less, and dropped. The test shall be repeated for the three other edges for a total of four drops. At the conclusion of the test, the frequency meter shall show no physical damage and shall comply with 3.4.

4.7.2.4 Vibration. - The frequency meter shall be securely packed in its commercial shipping container and subjected to a vibration whose frequency shall be varied linerally over the range from 5 to 52 cps with a double amplitude of 0.036 inches. The range shall be traversed from minimum to maximum frequency and returned to minimum frequency within one minute. The test shall be conducted for 15 minutes along each of three mutually perpendicular axes for a total of 45 minutes. At the conclusion of the test, the frequency meter shall show no physical damage and shall comply with 3.4.

4.7.2.5 Shock. - The frequency meter shall be securely packed in the commercial shipping container and dropped from a height of 3 feet on the bottom face, a bottom edge, and a bottom corner. At the conclusion of the test, the frequency meter shall show no physical damage and shall comply with 3.4.

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4.8 Sampling.

4.8.1 Environmental Tests. - The environmental tests of 4.7 shall be performed on a representative frequency meter as specified by the contracting officer.

4.8.2 Electrical Tests. - The electrical tests of 4.6 shall be performed on each frequency meter.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - The frequency meters shall be packaged, packed and marked in accordance with normal commercial preservation methods and shall be prepared for delivery to insure acceptance by common or other carrier for safe transportation to point of delivery. The preparation for delivery shall conform to the requirements of the Consolidated Freight Classification Rules in effect at the time of shipment.

6. NOTES

6.1 Intended Use. - The frequency meters covered in this specification are components of Microwave Calibration Test Sets AN/USM-129, AN/USM-129A and AN/USM-129B. They are incorporated in the Universal Microwave Calibrator panels specified in MIL-U-46351(ORD).

6.2 Ordering Data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Applicable stock number.

(c) Provisions for initial production sample, including date of delivery (see 3.1).

(d) Environmental tests required (see 4.8.1).

6.3 Design Information. - Design information regarding specific equipment not covered or referenced by this specification may be obtained from the procuring activity upon specific request of the contractor.

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